ME 200 Thermodynamics

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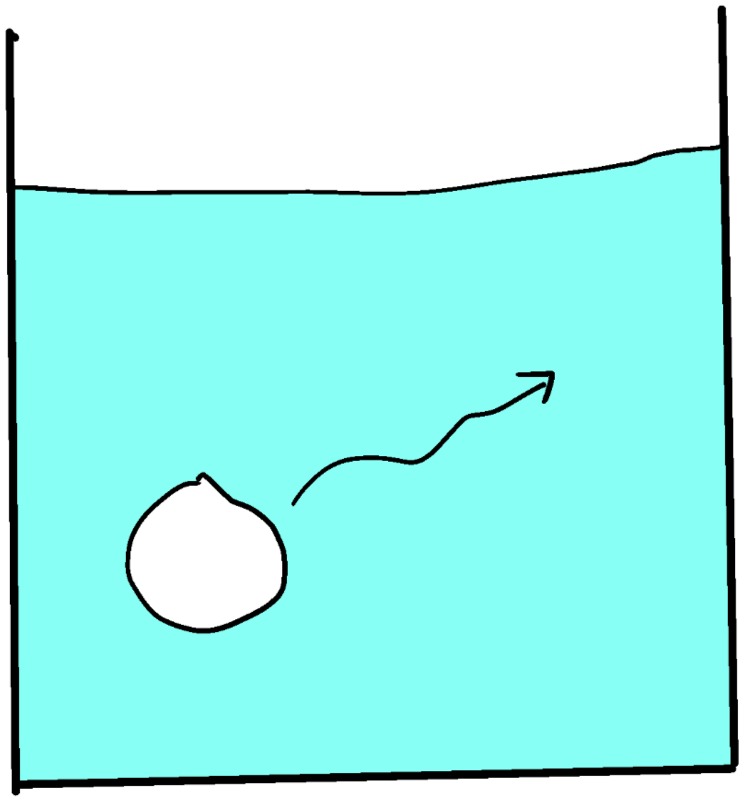
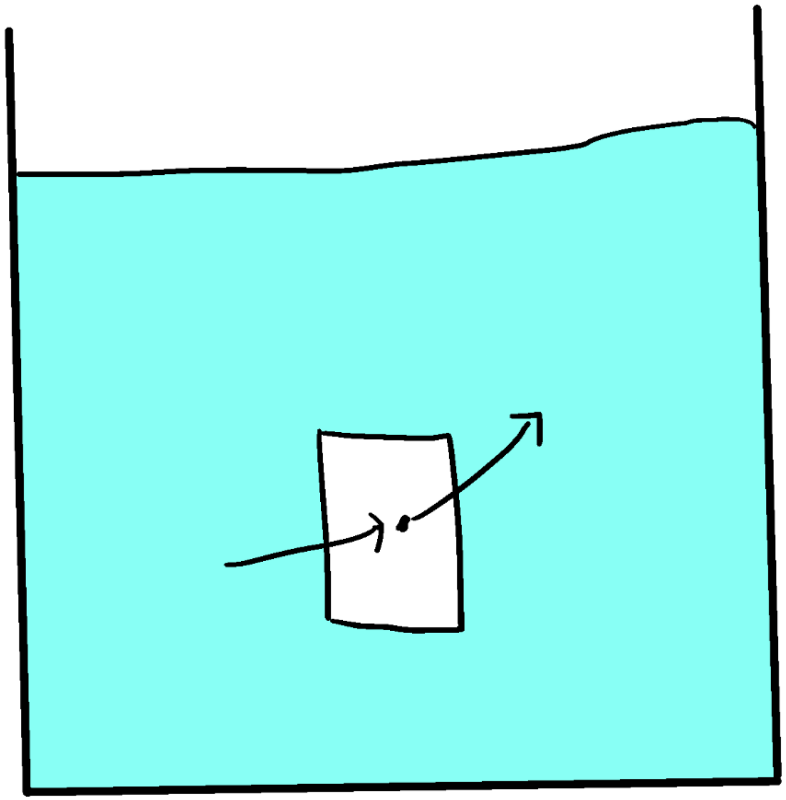
Lecture 02 – Properties & work

Systems



LAGRANGIAN



EULERIAN

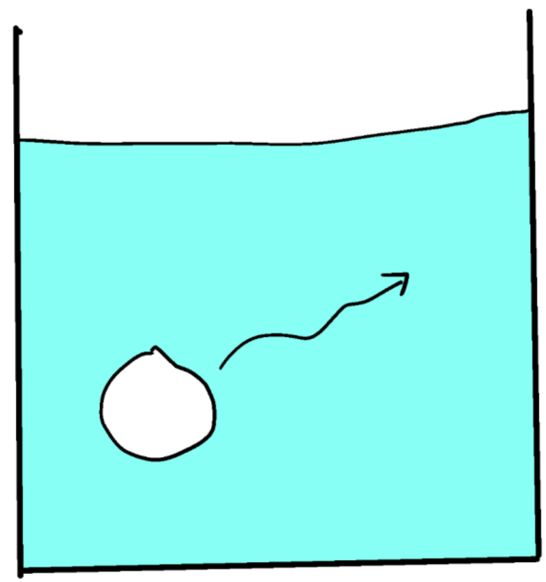
Systems

Closed System

It is a particular (unchanging) mass enclosed by a boundary (real or imagined, fixed or moving). Heat or work can cross the boundary, but mass cannot.

Applying our "laws" this way (to a particle or mass system) is a LAGRANGIAN view.

(Particle tracking)
(Control mass)



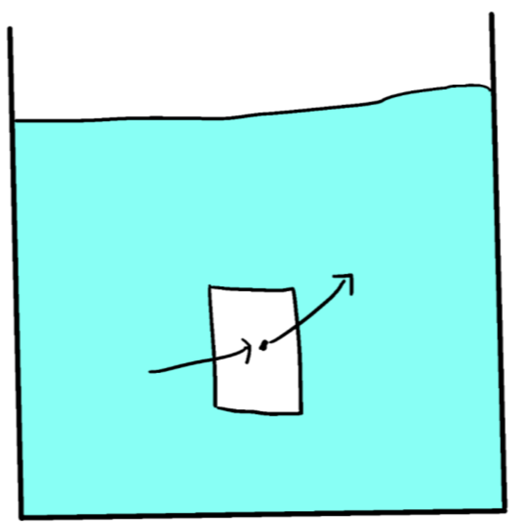
Systems

<u>Open System</u>

It is a particular volume, enclosed by a boundary (real or imagined, fixed or moving). Heat, work, <u>and mass</u> can cross the boundary.

Applying our "laws" this way (to a particular volume) is an EULERIAN view.

(Control volume)



Properties

Property : any quantity that describes the state of a system
any quantity whose value depends solely on the state of the system under study.
(totally independent of history or future)

Intensive properties – independent on system mass

P, T, v, u, h, s, x (quality)

Extensive properties – depends on system mass

V, U, H, S
U: internal energy; H: enthalpy; S: entropy

Pressure

Pressure – force applied perpendicular to the surface of an object per unit area over which that force is distributed

$$1 \text{ standard atmosphere (atm)} = \begin{cases} 1.01325 \times 10^5 \text{ N/m}^2 \\ 14.696 \text{ lbf/in.}^2 \\ 760 \text{ mmHg} = 29.92 \text{ inHg} \end{cases}$$

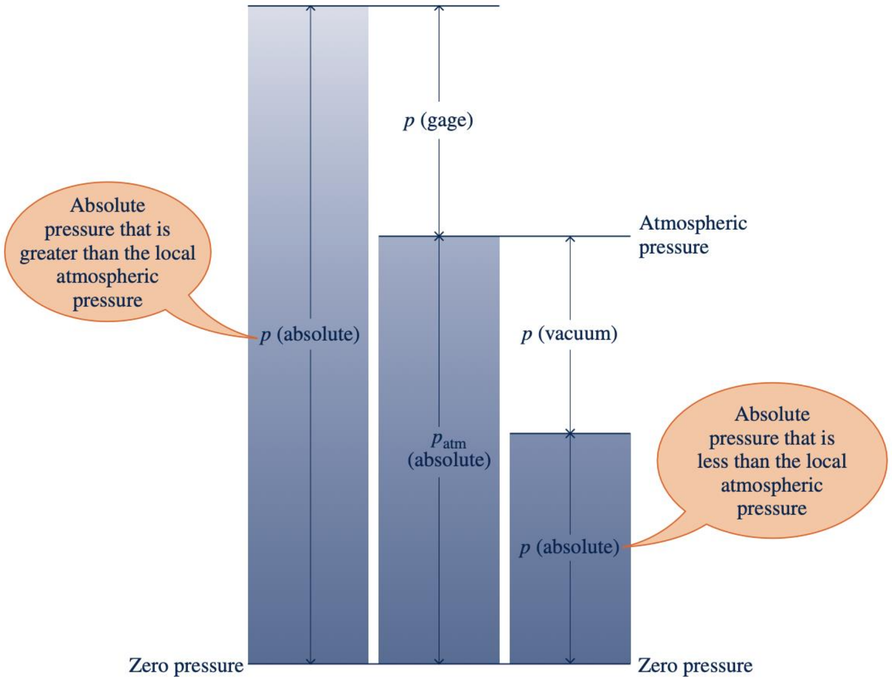
$$1 \text{ bar} = 10^5 \text{ N/m}^2$$

gage pressure :

$$p(\text{gage}) = p(\text{absolute}) - p_{\text{atm}}(\text{absolute})$$

vacuum pressure :

$$p(\text{vacuum}) = p_{\text{atm}}(\text{absolute}) - p(\text{absolute})$$



Temperature

Temperature (T) – measure of the average kinetic energy of the particles in a system

$$T(°R) = 1.8T(K)$$

$$T(°C) = T(K) - 273.15$$

$$T(°F) = 1.8T(°C) + 32$$

$$T(°F) = T(°R) - 459.67$$

Kelvin, Celsius, Fahrenheit, Rankine, etc.

Volume

Volume (V) – (extensive) [m^3]

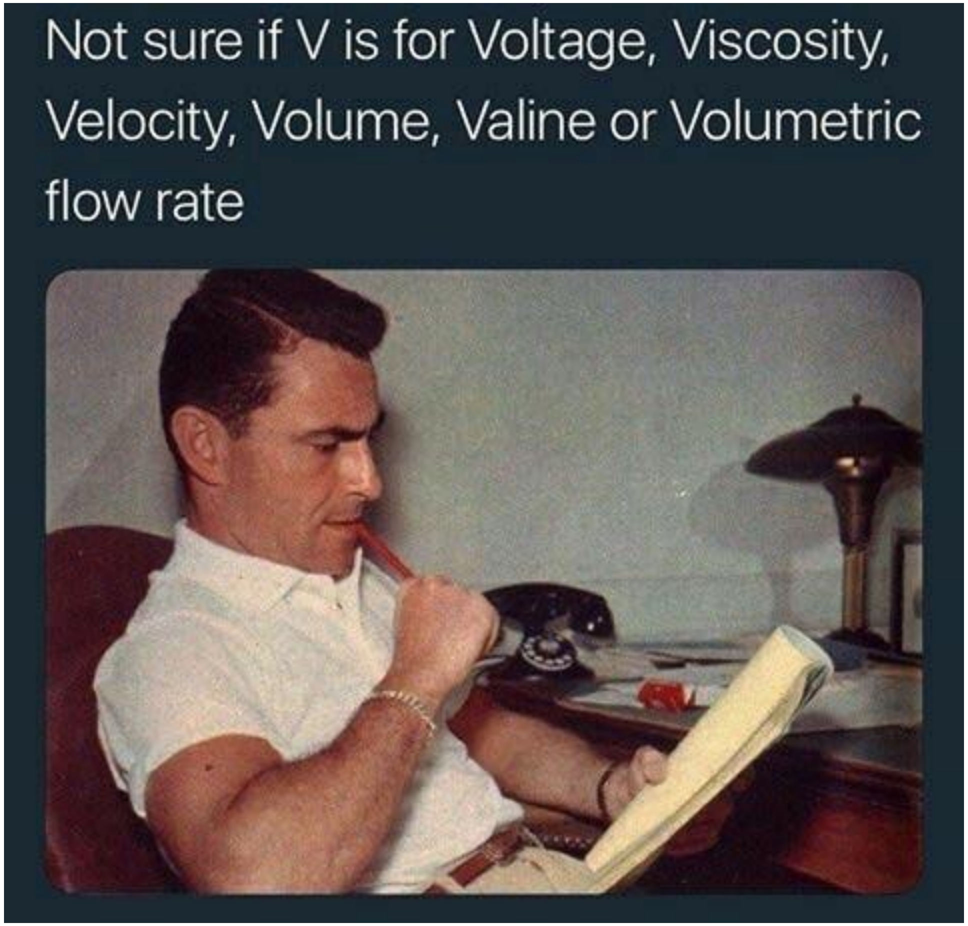
- Quantity of three-dimensional space occupied by an object or substance

Specific Volume (*v*) – (intensive) [m^3/kg]

- Number of cubic meters occupied by one unit of a particular substance

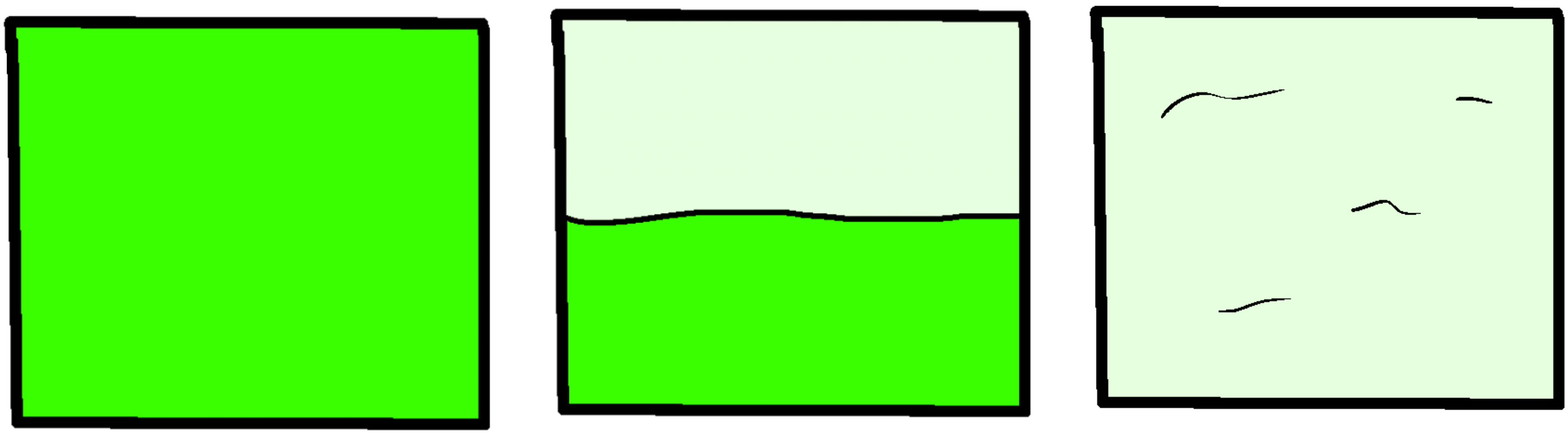
$$v = \frac{V}{m} = \rho^{-1}$$

Volume


Not sure if V is for Voltage, Viscosity, Velocity, Volume, Valine or Volumetric flow rate

Quality

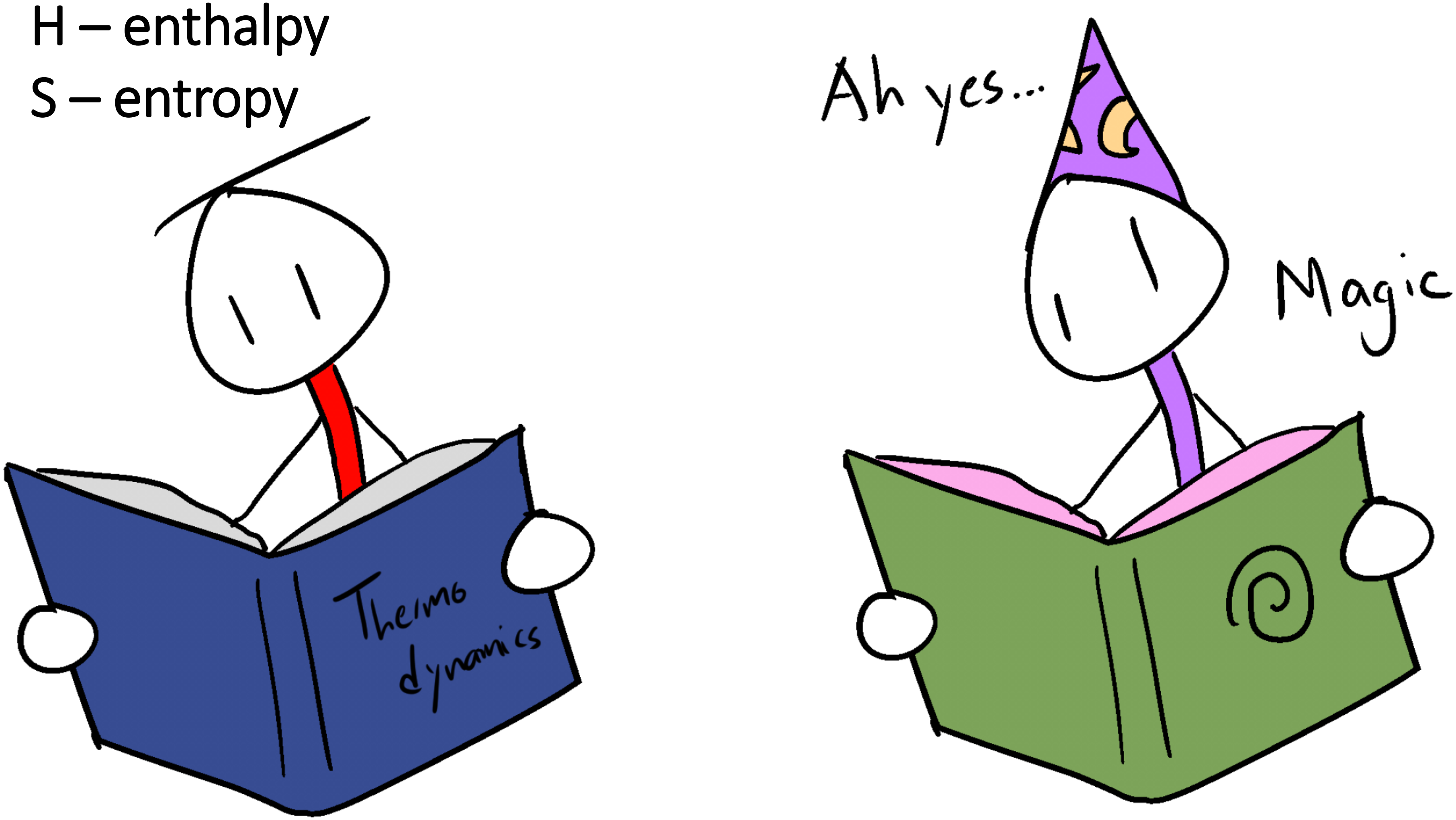
Quality (x) – vapor quality; the mass fraction in a saturated mixture that is vapor



$x = 0$ $x = 0.5$ $x = 1$

Properties

U – internal energy
H – enthalpy
S – entropy



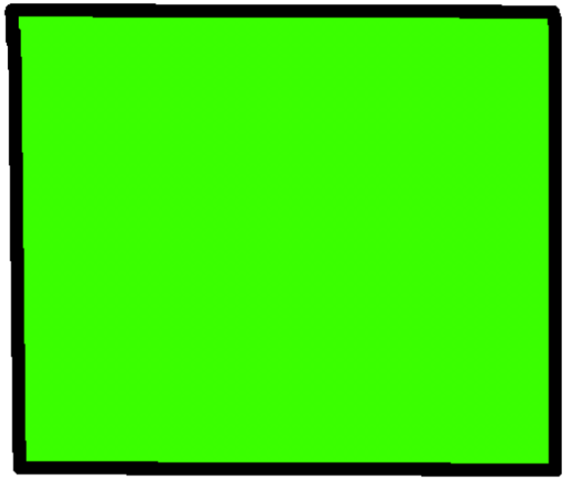
State

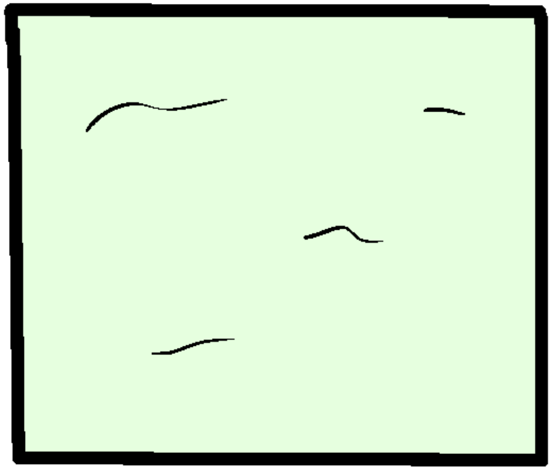
Usually, with 2 independent intensive properties, we can describe a state.

State: condition of system described by properties

ex) state 2, steam at p_2, v_2, T_2, u_2, h_2, s_2, etc

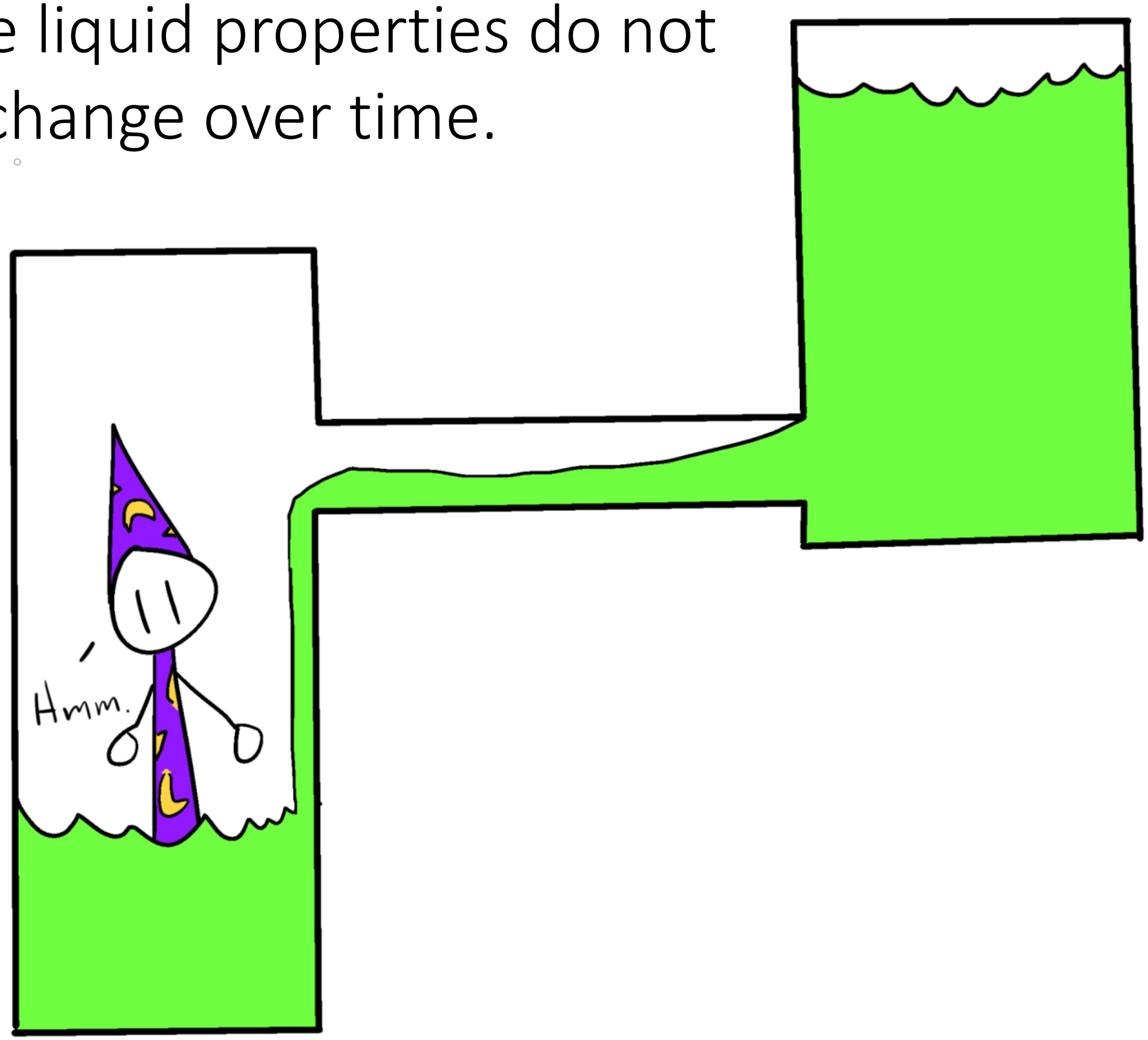
Steady-state: properties of system are unchanging over time





State

Mark a system where it is steady-state!
Assume liquid properties do not change over time.



Process

Process: transformation between 2 states

ex) Steam flow through turbine

→ a process from state 2 to state 3

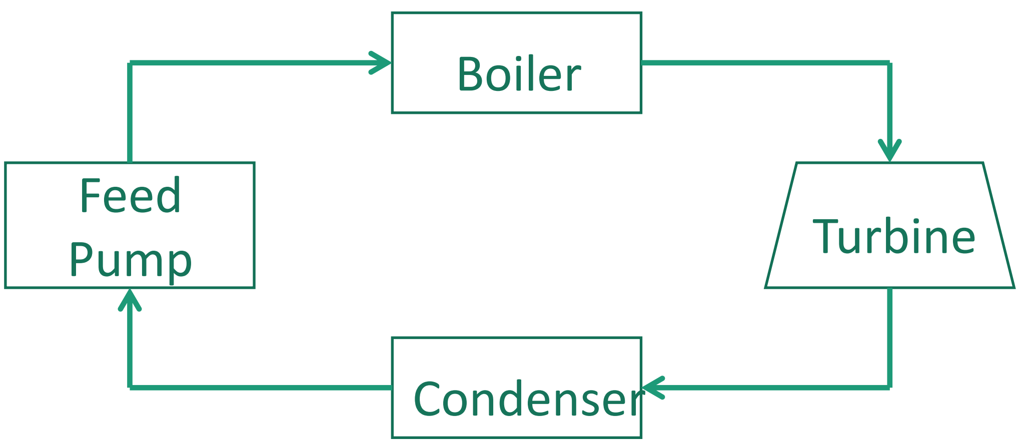
However,

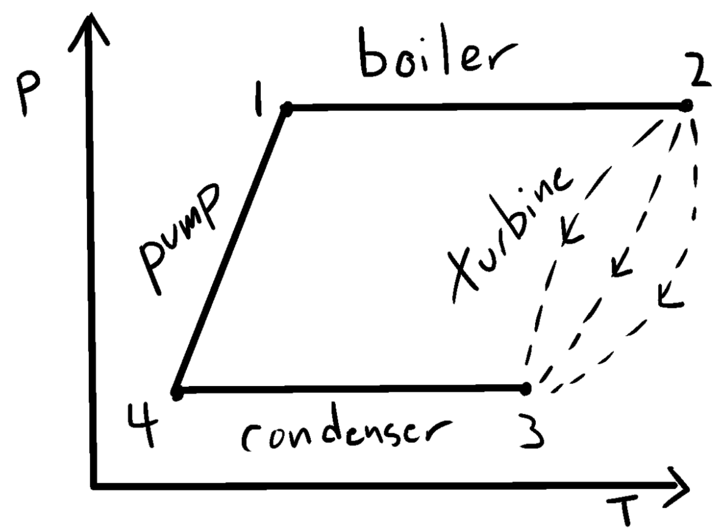
the change between properties is independent of the process

i.e. v_3-v_2 depends only on states 2 & 3

Work and Energy Transfer depend on processes

→ not properties





Cycle

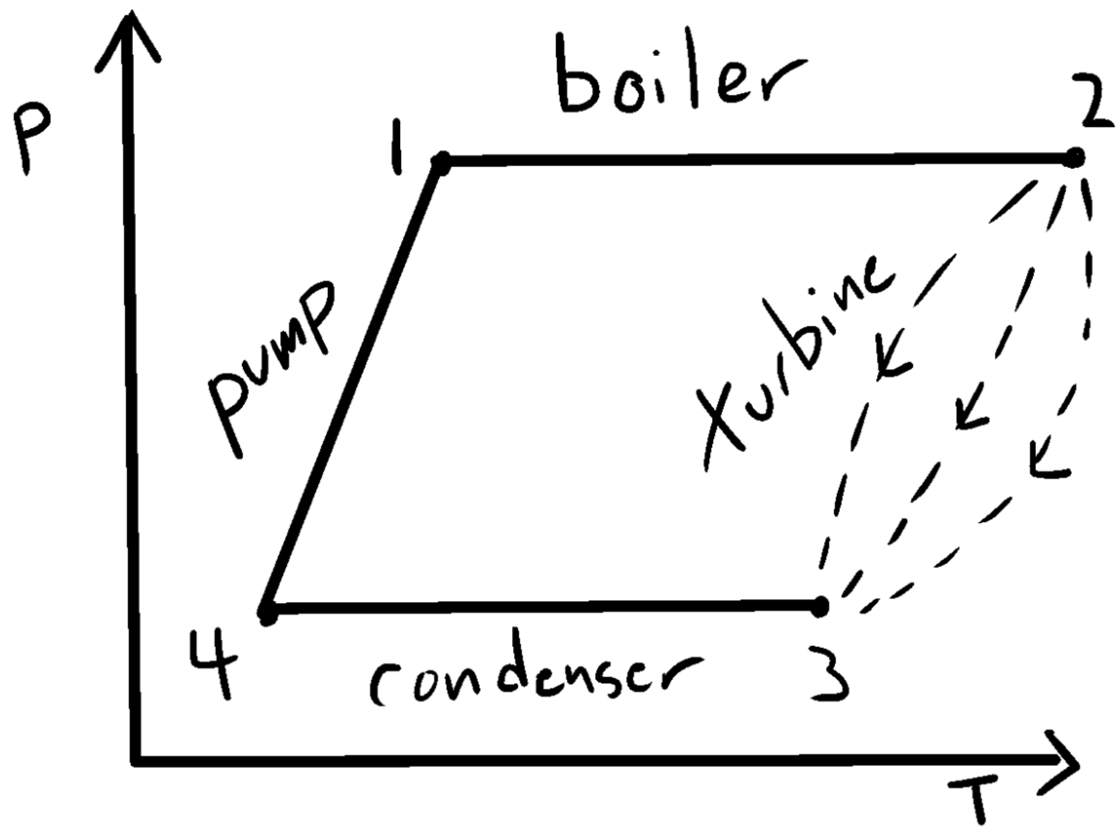
Thermodynamic Cycle: series of processes which begins & ends of same state

ex) 1→2→3→4→1

at 1, 2, 3, 4, all properties are known.

However, work outputs are different

from 3 processes

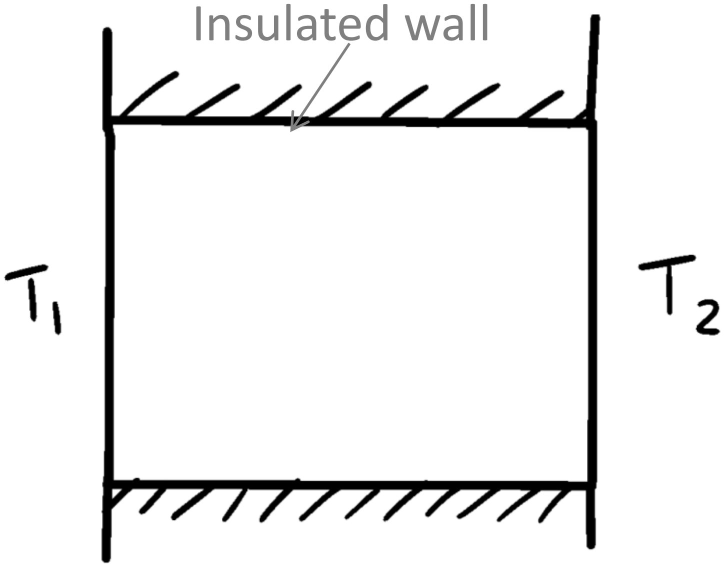


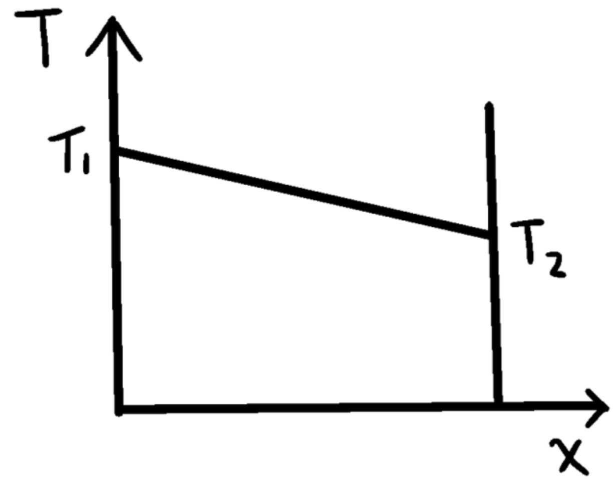
Equilibrium

Isolate a system from all surroundings and let it settle (relax) to a uniform state –

i.e. equilibrium state

Example: Hot, cold walls at surroundings

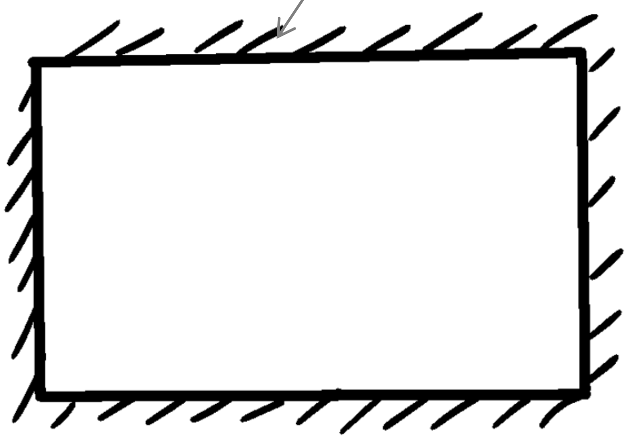


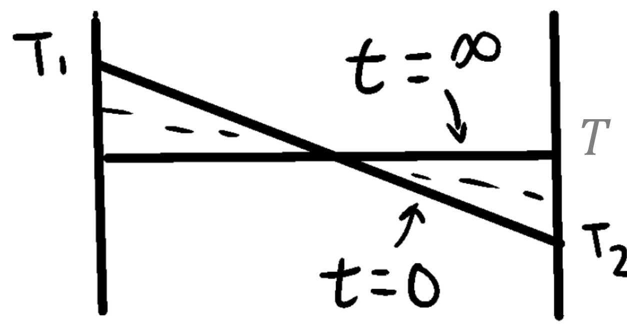


Steady-state, but not equilibrium since temperature is non-uniform.

Example: Isolate from surroundings (adiabatic)

Insulated on all sides, or “adiabatic” process.





$$T = \frac{1}{2}(T_1 + T_2)$$

The word **adiabatic** means *without heat transfer.* Thus, if a system undergoes a process involving no heat transfer with its surroundings, that process is called an *adiabatic process.*

Quasi-Equilibrium

During any real process, system generally is not in equilibrium.

<u>Quasi-equilibrium</u> process:

t_{eq} = characteristic time for system to settle from non-equilibrium state to equilibrium state.

t_{pr} = characteristic time for process

If $t_{eq} \ll t_{pr}$, process is so slow that system is always in equilibrium

Work

Work is equal to…

A. Base × Height
B. Force × Distance
C. Energy/Volume
D. Impulse × Velocity
E. Mass × Acceleration

Work

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A. Base × Height
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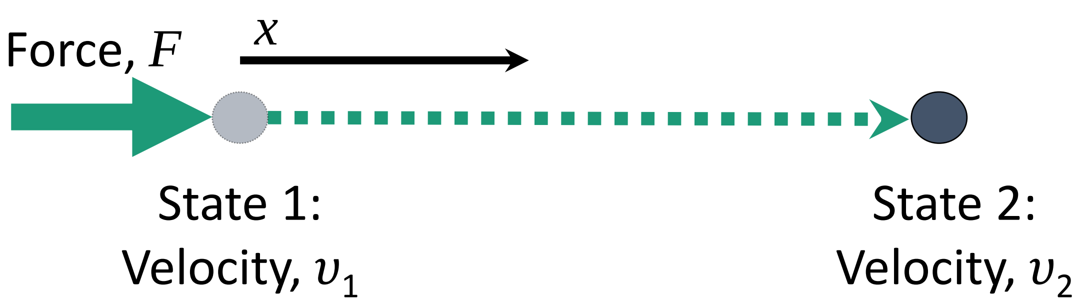
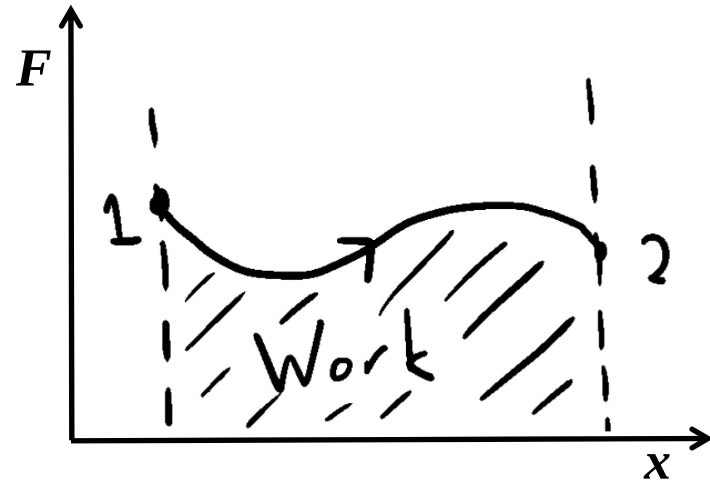
Work

Work = force × distance = energy transferred

Work depends on the path (process), so it is not a property

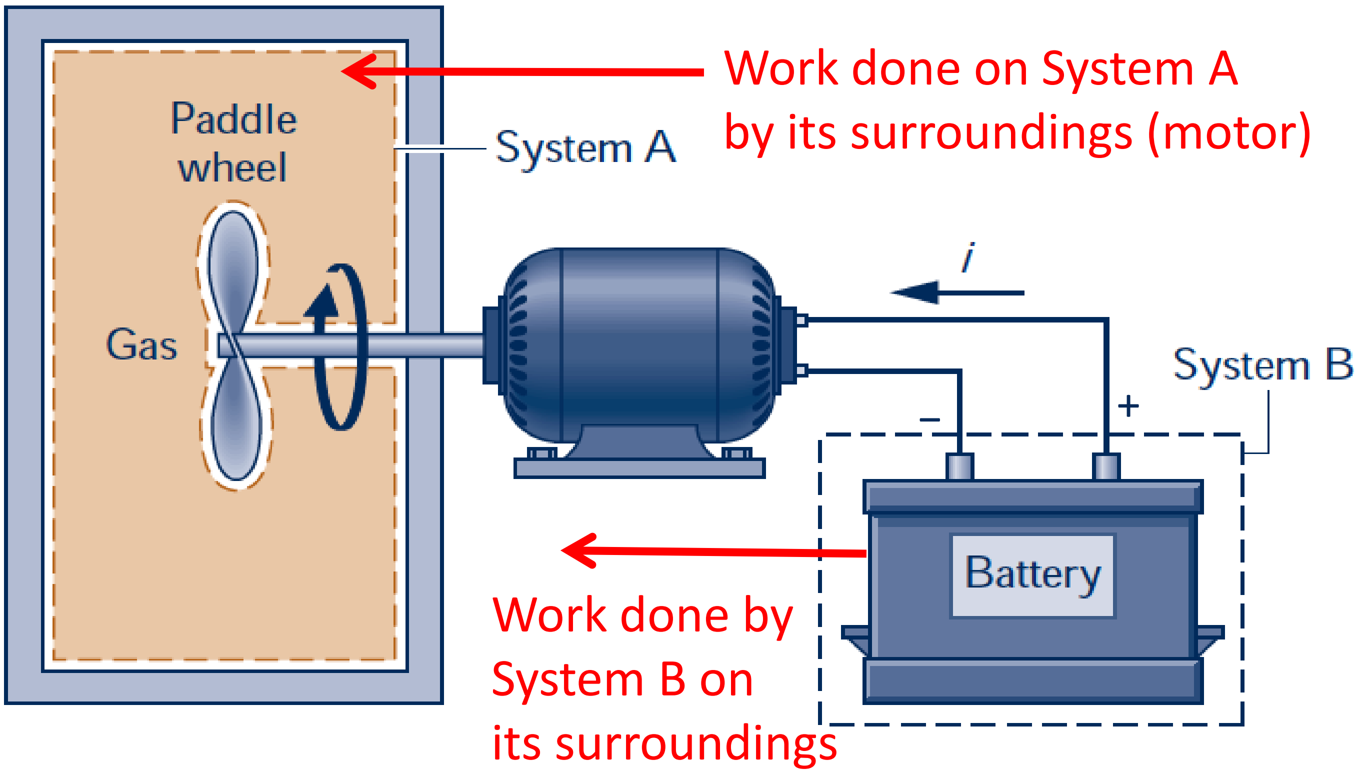




Thermodynamic Definition of Work

*Work is **done by** a system **on** its surroundings if the sole effect on everything external to the system could have been the raising of a weight.*

Essentially Force × Distance



Work Directionality and the Sign Convention

Work

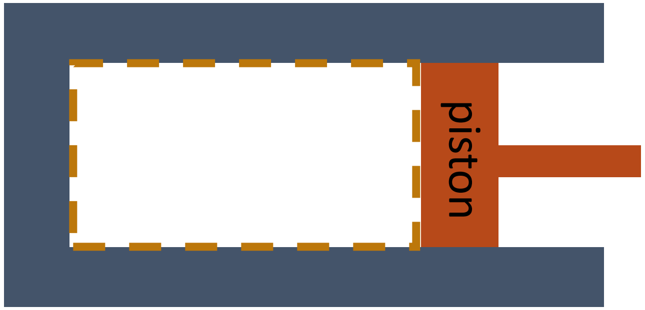
If Work, W, is done BY a system…

A. $W > 0$

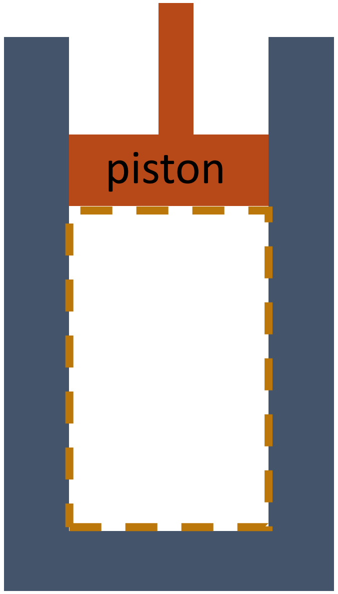
B. $W < 0$

Work done by a system on its surroundings is considered as positive – this is the convention used in Thermodynamics and thermal science classes.
This is consistent with the Work equation shown.

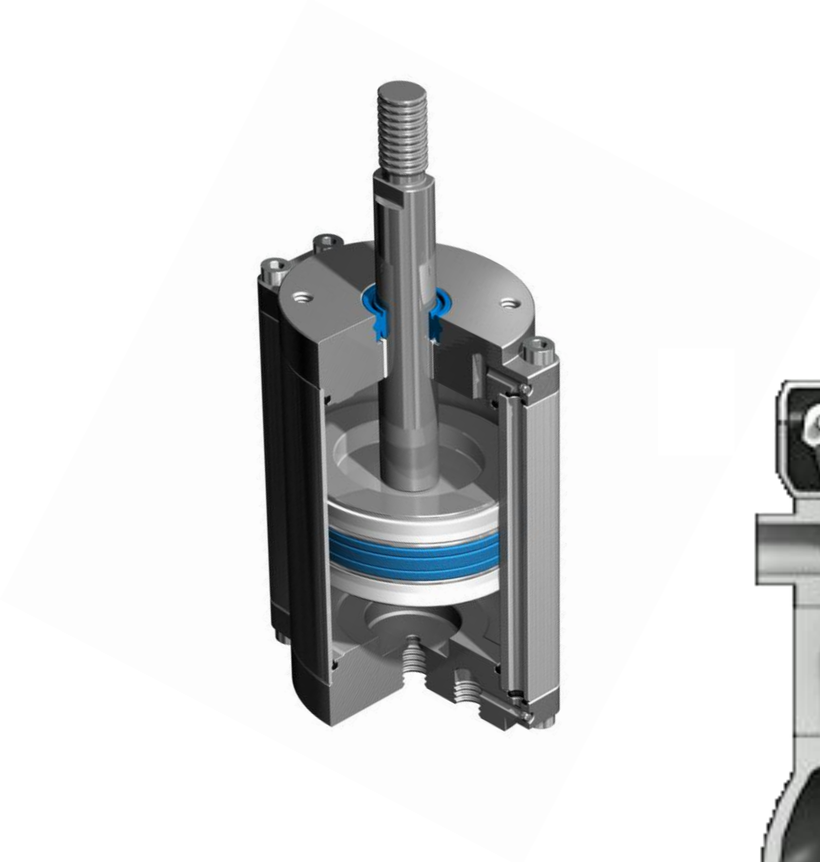
Piston-Cylinder Assembly

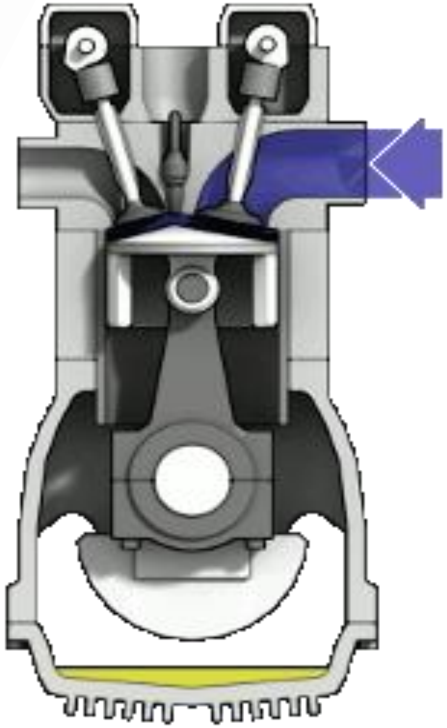


horizontal



vertical





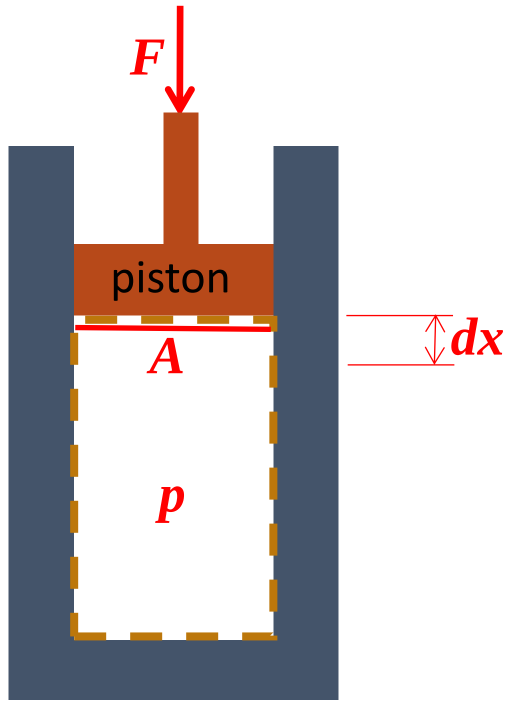
Work

Example: Piston work, i.e. p-V work

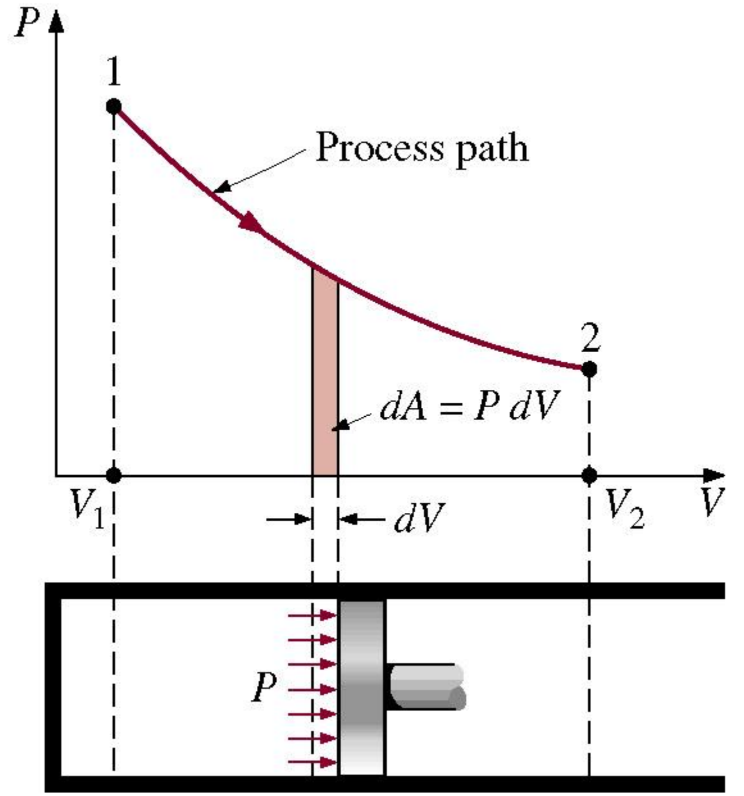
P=F/A

dV = A·dx = d(A·x)

$$W_{1-2} = \int_1^2 \mathrm{F}dx = \int_1^2 \mathrm{P}dV$$



Example: Constant Pressure Process at 5 MPa has a volume change from 1 m^3 to 10 m^3.

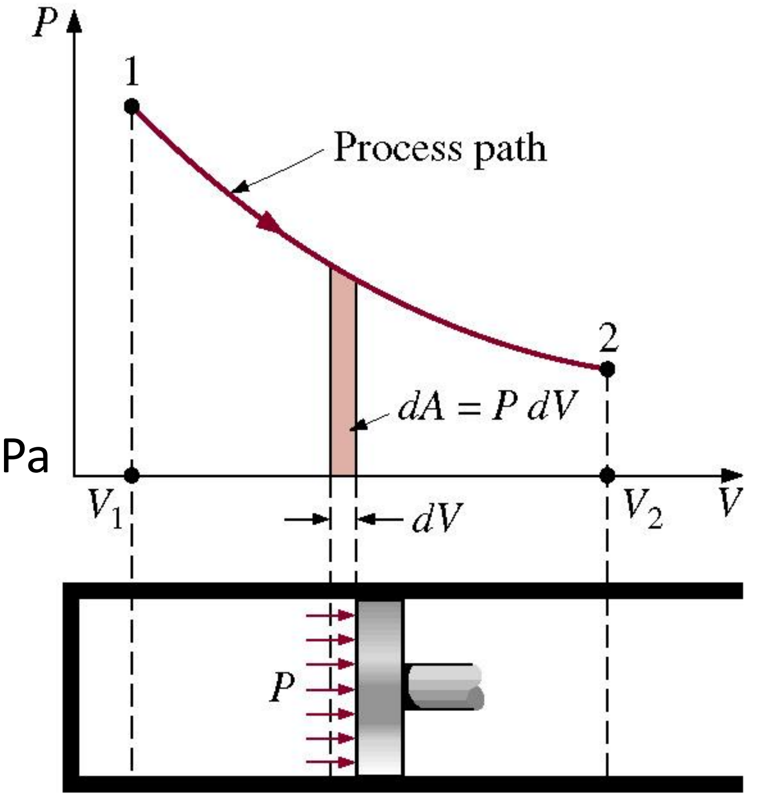


How to get a
Constant Pressure Process?

Work

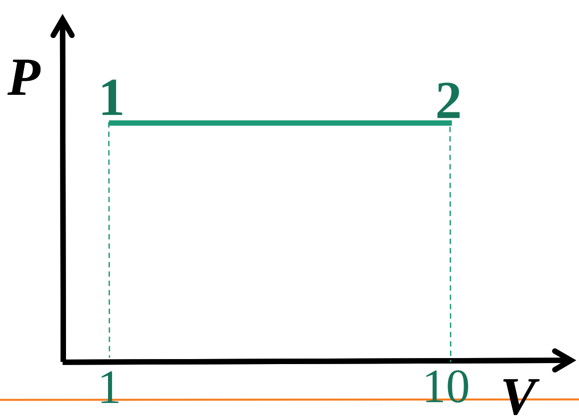
$$W = \int_{V_1}^{V_2} p dV$$

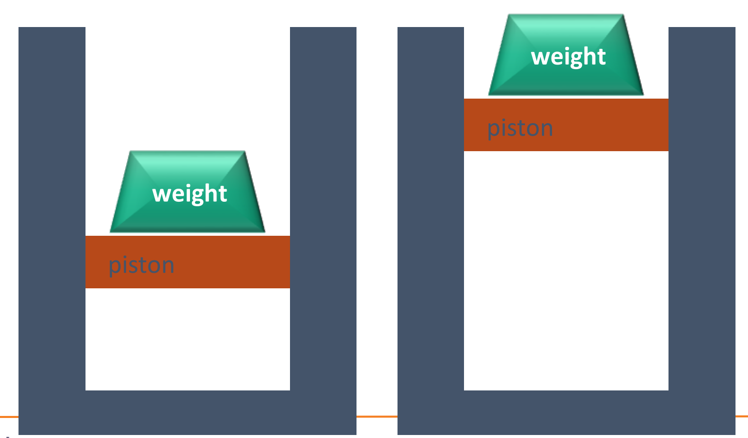
$$w = \frac{W}{m} = \int_{v_1}^{v_2} p dv$$



Example: Constant Pressure (Isobaric) Process at 5 MPa has a volume change from 1 m³ to 10 m³.

$$W = \int_{1}^{10} p dV = (5\ MPa)(10 - 1)m^3$$
$$= +45 \times 10^6\ N \cdot m\ (or\ J)$$





Ideal Gas

$$pV = n\bar{R}T = m\frac{n}{m}\bar{R}T = m\frac{1}{M}\bar{R}T = mRT$$

$$pV = mRT \Rightarrow pv = RT$$

Universal Gas Constant

$\bar{R}$ 8.314 kJ/kmol·K

1.986 Btu/lbmol·°R

Values of the Gas Constant *R* of Selected Elements and Compounds

Substance	Chemical Formula	*R* (kJ/kg·K)	*R* (Btu/lb·°R)
Air	—	0.2870	0.06855
Ammonia	NH_3	0.4882	0.11662
Argon	Ar	0.2082	0.04972
Carbon dioxide	CO_2	0.1889	0.04513
Carbon monoxide	CO	0.2968	0.07090
Helium	He	2.0769	0.49613
Hydrogen	H_2	4.1240	0.98512
Methane	CH_4	0.5183	0.12382
Nitrogen	N_2	0.2968	0.07090
Oxygen	O_2	0.2598	0.06206
Water	H_2O	0.4614	0.11021

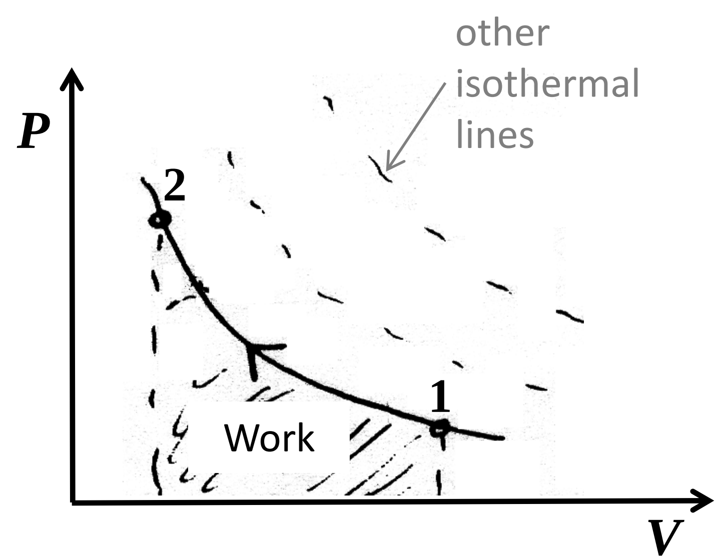
Work

Example: Isothermal Process

$$pV = mRT = constant \Rightarrow p = \frac{mRT}{V} = \frac{constant}{V}$$

$$W = \int_1^2 pdV = mRT \int_1^2 \frac{dV}{V} = mRT(lnV_2 - lnV_1)$$

$$= mRTln\left(\frac{V_2}{V_1}\right) = p_1 V_1 ln\left(\frac{V_2}{V_1}\right)$$

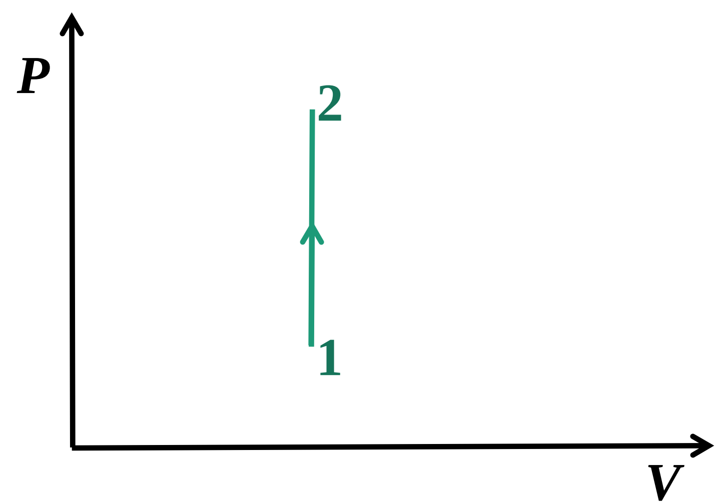


Example: Polytropic Process

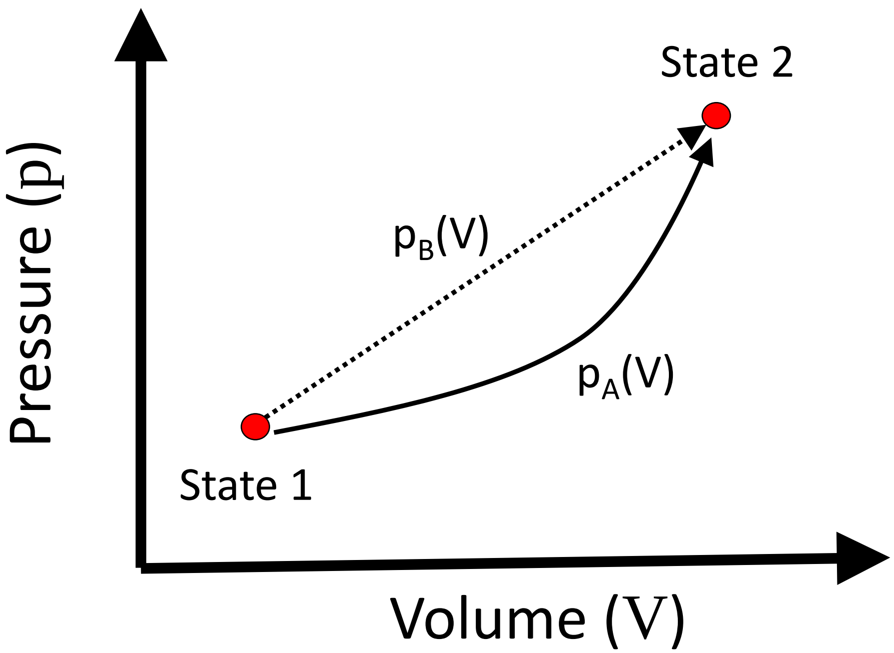
$$pV^n = constant$$

Example: Constant Volume Process

$$W = \int_1^2 pdV = 0 \quad \text{since} \quad dV = 0$$



Work



A. $W_{pV,A} > W_{pV,B} > 0$

B. $W_{pV,A} < W_{pV,B} < 0$

C. $0 < W_{pV,A} < W_{pV,B}$

D. $W_{pV,A} < 0 < W_{pV,B}$

E. $W_{pV,A} = W_{pV,B}$